Assessing Roadway Network Risk to Compound Flooding in Gulf Coast of Texas: An Integrated Hydrodynamic and Machine Learning Approach

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ABSTRACT

Coastal transportation networks are increasingly sensitive to compound flooding, where storm surge and heavy rainfall occur concurrently. Traditional flood risk assessments often analyze these hazards in isolation, potentially underestimating the true systemic risk to critical infrastructure. This study presents an integrated framework to quantify and classify roadway network risks in Galveston County, Texas, a region highly susceptible to hurricane impacts. A coupled 1D/2D hydrodynamic model (HEC-RAS) was developed and validated against data from 2008 Hurricane Ike, achieving a robust model fit (R^2=0.62 and KGE=0.66) with observed high-water marks. The validated model was then used to simulate a 100-year, 24-hour compound flood event, revealing widespread inundation driven by the interaction of surge and rainfall-runoff. To move beyond binary hazard mapping, the study employed an unsupervised machine learning approach to partition the roadway network based on physical exposure, operational criticality, and socio-economic vulnerability. The analysis identified four statistically distinct risk typologies, ranging from "Minor" to "Catastrophic." Key findings reveal a "Catastrophic Failure" cluster that isolates the region's logistical spine where high traffic volumes intersect with extreme flood depths, and a "Major Risk" cluster within the urban grid representing a "double jeopardy" zone of high physical hazard and social immobility. These results demonstrate that roadway vulnerability is driven as much by socio-economic geography as by physical factors, providing a data-driven basis for prioritizing equitable resilience investments.

Keywords: Compound Flooding, Roadway Risk, HEC-RAS, K-Means Clustering, Coastal Resilience, Transportation Infrastructure, Hurricane Ike.

1. INTRODUCTION

The increasing frequency and intensity of extreme weather events pose a significant and escalating threat to coastal communities and their critical infrastructure worldwide (Hallegatte et al., 2013; Heberger et al., 2011; Khanam et al., 2021; Koetse & Rietveld, 2009; Nazarnia et al., 2020; Shen et al., 2019; Vousdoukas et al., 2018). Coastal regions are particularly susceptible to the complex dynamics of compound flooding, where the concurrent impacts of storm surge, high tides, and extreme precipitation interact to create inundation scenarios far more severe than any single hazard would in isolation (Bakhtyar et al., 2020; Khanam et al., 2021; Moftakhari et al., 2017; Pandey et al., 2021; Shen et al., 2019; Wahl et al., 2015). Recent research has identified these complex interaction areas as the Upper Transition Zone (UTZ), where the convergence of downstream coastal water levels and upstream riverine discharge creates a multi-faceted flood hazard that challenges traditional assessments (Mita et al., 2025). The Texas Gulf Coast is a prime example of a low-lying, densely populated area frequently exposed to hurricanes that generate such compound flooding. Events like 2008 Hurricane Ike have demonstrated that the interaction between inland rainfall-runoff and coastal storm surge can create complex flooding dynamics that are not captured by traditional, single-hazard floodplain maps (Green et al., 2025; Ray et al., 2011).

To understand the complex hazards of compound flooding, researchers have increasingly turned to advanced hydrodynamic models. The scholarly literature documents a clear progression from one-dimensional (1D) models, suitable for in-channel flow, to two-dimensional (2D) and coupled 1D/2D approaches that more realistically simulate the multi-directional flow of water across broad, complex floodplains and urban environments (Dimitriadis et al., 2016; Horritt & Bates, 2002; Jibhakate et al., 2022; Pasquier et al., 2019; Quirogaa et al., 2016; Shaikh et al., 2023; Xu et al., 2023). While 1D models are computationally efficient, they often fail to capture momentum exchanges and overland flow propagation in wide, complex floodplains (Bakhtyar et al., 2020; Pasquier et al., 2019). The accuracy of these models, however, is highly sensitive to the resolution of input data, particularly topographic Digital Elevation Models (DEMs) and land cover-derived surface roughness coefficients, highlighting the trade-offs between data acquisition costs, computational demand, and model fidelity (Afshari et al., 2018; Di Salvo et al., 2018; Purvis et al., 2008; Wadey et al., 2015; Yalcin, 2020). Consequently, robust flood hazard assessment now often involves coupling hydrologic models to generate runoff boundary conditions for 2D hydraulic models, thereby creating a comprehensive modeling cascade that translates hydroclimatic inputs into high-resolution inundation maps (Bates et al., 2021; Knebl et al., 2005; Najafi et al., 2021; Pandey et al., 2021; Zhang et al., 2022).

However, producing an accurate inundation map is only the first step in a comprehensive risk analysis. Recent research highlights that holistic assessments must quantify risk as a multi-faceted concept—encompassing physical exposure, intrinsic infrastructure characteristics, and socio-economic conditions—shifting the focus of flood impact analysis from direct physical damage toward understanding systemic disruptions within and between critical infrastructure networks(Abenayake et al., 2022; Najafi et al., 2021; Pregnolato et al., 2017). Recent studies have repeatedly demonstrated that the disruption of transportation systems, in particular, leads to significant cascading or "knock-on" effects, such as widespread traffic delays, economic losses from business interruption, and compromised access to emergency services (Borowska-Stefańska et al., 2025; Pyatkova et al., 2019; Salvo et al., 2025). This systemic perspective reveals that the criticality of an infrastructure component is not merely a function of its physical exposure, but also its topological importance within the network and the interdependence of the systems it supports (Lu & Peng, 2011; Uzcategui-Salazar & Lillo, 2022).

In response to this complexity, risk assessment methodologies have become increasingly sophisticated. A critical validation gap persists in literature; while many studies simulate theoretical design storms, fewer rigorously validate hydrodynamic models against ground-based High-Water Marks (HWMs) collected during historical compound events (Saleh et al., 2017). Furthermore, while frameworks for assessing the risk of interdependent systems are emerging (Martello et al., 2021), a gap remains in applying these concepts at a local scale through the integration of high-resolution hydrodynamic modeling with data-driven classification methods. Methodologies such as multi-criteria analysis and machine learning (ML) clustering, for example, have proven effective in synthesizing diverse datasets to create holistic risk assessments that can inform targeted adaptation strategies (Bounaceur et al., 2022; Eini et al., 2020; Javadi et al., 2017; Luo et al., 2024). This approach allows for an objective, evidence-based identification of risk hotspots that might be overlooked by traditional, hazard-centric analyses. Consequently, there remains a critical need for local-scale frameworks that combine rigorously validated hydrodynamic modeling with multi-dimensional data integration to move beyond simplified hazard assessments.

This study addresses these gaps by presenting a methodology to identify the risky roadway network hotspots in Galveston County, Texas. The specific objectives of this research are: (1) to construct a high-resolution coupled 1D/2D HEC-RAS model that explicitly accounts for the interaction between storm surge and rainfall-runoff and validate the model's performance against observational data from the 2008 Hurricane Ike event; (2) to quantify the spatial extent and depth of compound flooding under a 100-year design scenario; and (3) to identify the susceptible roadways using ML technique. The goal is to move beyond simple inundation mapping toward a more holistic risk analysis by integrating these physical characteristics with the intrinsic properties of road infrastructure and the socio-economic context of surrounding communities. By coupling operationally standard hydrodynamic modeling—utilizing excess rainfall inputs from HEC-HMS—with an unsupervised machine learning classification approach, this research provides a nuanced, spatially explicit risk map to assist decision-makers in prioritizing resilience-focused investments and improving emergency management planning.

2. DATA AND METHODOLOGY

Galveston County, situated on the Texas Gulf Coast, serves as the study area for this research (Figure 1) which illustrates the county's low-lying coastal setting, the major roadway inventory, and its proximity to Galveston Bay. As a key component of the Houston-The Woodlands-Sugar Land metropolitan area, the county's dense transportation network is critical for regional mobility, economic activity, and emergency evacuation. Its geography is characterized by extremely low-lying coastal plains, barrier islands (such as Galveston Island and Bolivar Peninsula), and a network of bayous that drain into the Galveston Bay estuary system. The county's average elevation is only around 3 meters (10 feet) above sea level, a topography that makes it exceptionally risky to hurricane- induced storm surge and intense inland rainfall (Ray et al., 2011).

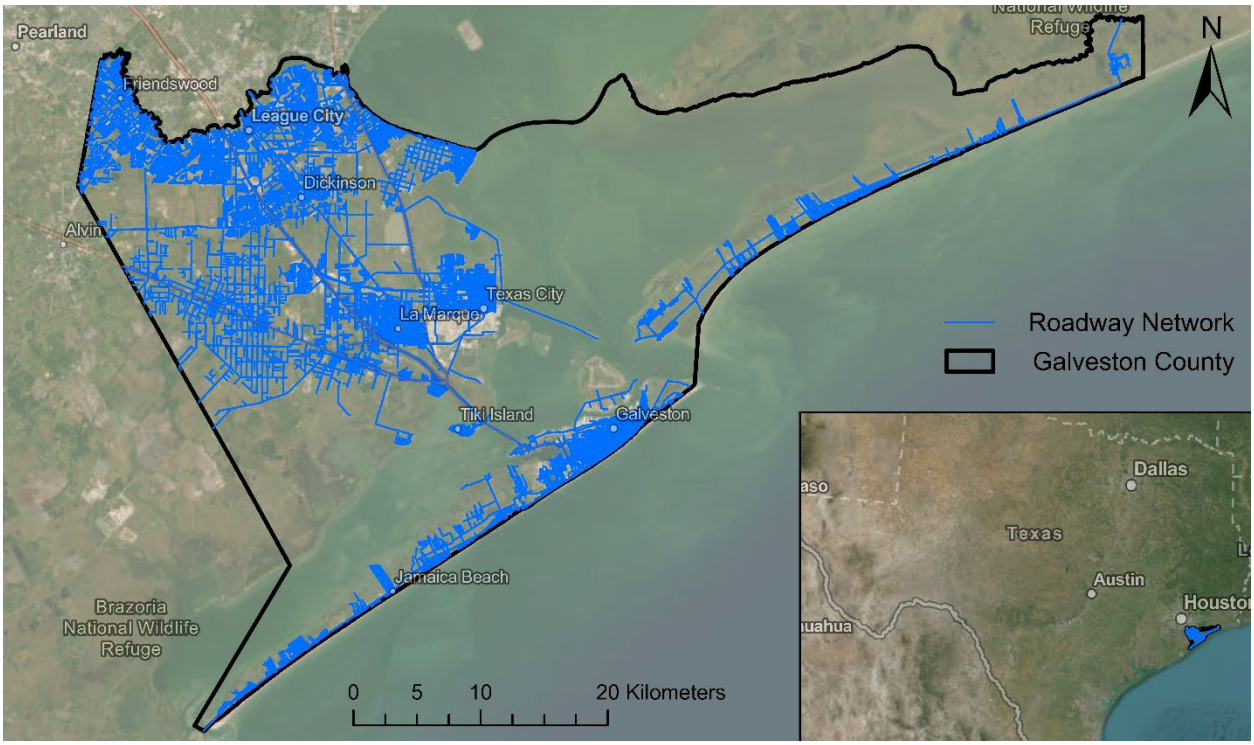

Figure 1. Study area encompassing Galveston County, Texas, highlighting the roadway network used for the flood vulnerability assessment. The county boundary and all roadway segments included in the analysis are shown in blue.

The National Oceanic and Atmospheric Administration (NOAA) maintains a dense network of real-time sensors in the US coasts and Great Lakes, including at the Galveston Bay, as part of its Physical Oceanographic Real-Time System (PORTS®), which provides critical data essential for both maritime operations and flood modeling (*NOAA Tides and Currents*).

The selected region is frequently subjected to tropical cyclones, which pose a substantial and recurring threat to the county's population of over 360,000 residents (*Census Bureau Tables, 2023*). Hurricane Ike in 2008 produced a devastating storm surge that reached up to 6.7 meters (22 feet) at Sabine Pass and averaged near the 100-year flood levels across the county, inundating an estimated 100,000 homes in Texas and causing billions of dollars in damage (Berg, 2009). This history of catastrophic events, combined with projections of accelerated sea-level rise and increased potential for extreme precipitation, necessitates a detailed analysis of the roadway network's risks to compound flooding (Heberger et al., 2011).

2.1 Data

The integrated flood risk assessment presented in this study relies on a suite of geospatial, hydrological, hydraulic, and socio-economic datasets. Each dataset was acquired from authoritative sources and pre-processed to meet the specific requirements of the modeling and analysis framework. The primary topographic data source for the study area was a 3-meter resolution Digital Elevation Model (DEM) from the U.S. Geological Survey (USGS). High-resolution topographic data is a critical component for accurate 2D hydrodynamic modeling, as it directly governs the simulation of overland flow paths, water storage, and inundation extent, particularly in low-gradient urban and coastal environments (Afshari et al., 2018; Yalcin, 2020).

To characterize surface friction, the 2023 National Land Cover Database (NLCD) was used to assign spatially distributed Manning's *n* roughness coefficients, a standard approach for enhancing the physical realism of 2D hydraulic models (Afshari et al., 2018). The NLCD was also instrumental in defining spatially variable friction slopes for the normal depth boundary conditions within HEC-RAS.

Hydrologic and hydraulic model inputs were derived from several key national sources. The design precipitation for the 100-year, 24-hour storm event was sourced from NOAA Atlas 14, which provides authoritative rainfall frequency estimates (Perica et al., 2018). To translate this rainfall into runoff for the HEC-HMS model, Curve Numbers (CNs) were calculated based on soil characteristics from the Gridded Soil Survey Geographic (gSSURGO) database provided by the United States Department of Agriculture (USDA) (*Gridded Soil Survey Geographic (gSSURGO) Database | Natural Resources Conservation Service*). The crucial downstream boundary condition, representing the coastal water level, was developed using historical data from NOAA's Tides and Currents database (*NOAA Tides and Currents*). This allowed for a flood frequency analysis of storm tides, ensuring the 100-year coastal water level was realistically represented. To validate the hydrodynamic model's performance, simulated results from a hindcast of 2008 Hurricane Ike were compared against post-storm high-water marks collected by the Federal Emergency Management Agency (FEMA) (*Federal Emergency Management Agency*, 2009). Hurricane Ike was selected as the calibration and validation case study due to its status as a benchmark hydrometeorological event for the upper Texas coast, characterized by an exceptionally expansive wind field and catastrophic storm surge. Unlike typical storms where surge is primarily driven by peak wind intensity near the eye, Ike's surge was driven by a massive wind field that pushed water into the coast well before the center made landfall, resulting in inundations of 15–20 feet on the Bolivar Peninsula and 10–15 feet on Galveston Island (Berg, 2009). This complex surge generation mechanism, which produced water levels comparable to Category 4 storms despite Ike's Category 2 landfall intensity, provides a rigorous stress test for the hydrodynamic model's ability to capture wind-wave-surge interactions (*NOAA Water Level and Meteorological Data Report*, 2009). Furthermore, the event is extensively documented, with high-fidelity post-storm datasets—including comprehensive high-water marks (*Federal Emergency Management Agency*, 2009), continuous tide gauge records, and coastal change lidar surveys (Doran et al., 2009)—providing the necessary ground truth to robustly evaluate the model's accuracy.

The risk analysis was built upon the integration of infrastructure and socio-economic data within a GIS environment. The roadway network, including the spatial location and functional classification of all roads in Galveston County, was obtained from the Texas Department of Transportation (TxDOT) 2023 roadway inventory (*TxDOT Roadway Inventory*2023). This foundational infrastructure layer was then enriched with data from the Smart Location Database (SLD), a nationwide dataset that provides comprehensive socio-economic and built environment indicators at the census block group level. This multi-criteria approach provides a holistic view of risk, acknowledging that risk is a function of more than just physical flood exposure (Anand & Marcotullio, 2024; Clarke et al., 2024). A summary of all datasets used in this study is provided in Table 1.

Table 1. Summary of datasets used in this study, including data type, source agency, spatial and temporal resolution, data collection year, and role in the flood-risk modelling framework (Input, Auxiliary, or Validation). The datasets comprise topographic, meteorological, land-related, coastal, flood, transportation, and socio-economic components required to characterize flood drivers, surface hydrodynamics, roadway exposure, and community vulnerability in Galveston County, TX.

Type	Data	Source	Spatial Resolution	Temporal Resolution	Data Collection Year	Remarks
Topographic Data	Projection File	Spatial Reference	NA	Static	NA	Input
	Digital Elevation Model (DEM)	USGS	3 m	Static	2007	Input
Meteorological Data	Design Precipitation	NOAA ATLAS 14	NA	Hourly	NA	Input
	Historical Precipitation	NOAA NCEI	NA	Hourly	1946-1963 and 1993-2024	Auxiliary
Land Related Data	Land Use and Land Cover (LULC)	NLCD	30 m	Static	2023	Input
	Soil Group	gSSURGO	NA	Static	2022	Auxiliary
	Impervious	NLCD	30 m	Static	2023	Auxiliary
	Curve Number (CN)	NLCD	NA	Static	NA	Input
	Manning's n	NLCD	NA	Static	NA	Input
Coastal Data	Sea Water Level	NOAA Tides and Currents	NA	Hourly	1904 to 2024	Input
	Tidal Datums and Tidal Flooding	NOAA Tides and Currents	NA	Hourly	1900 to 2024	Auxiliary
	Storm Surge	NOAA Tides and Currents	NA	Hourly	1901 to 2024	Auxiliary
GIS	County Boundary	City of Galveston	NA	Static	2022	Input
Flood Related Data	Depth	FEMA	NA	sStatic	2008	Validation
Transportation Data	Roadway Inventory	TxDOT	NA	Static	2023	Input
Socio-economic Data	Socio-economic Variables	SLD	NA	Static	2021	Input

2.2 METHODOLOGY

This study employs a multi-stage, integrated methodology designed to quantify compound flood hazards and classify roadway network risk in Galveston County, Texas. The framework synthesizes advanced techniques from hydrology, hydrodynamics, geospatial analysis, and machine learning to create a comprehensive, data-driven assessment. The approach proceeds sequentially through four interdependent stages: (1) data acquisition and curation, which establishes the geospatial and parametric foundation for the models; (2) hydrologic modeling using HEC-HMS to translate design precipitation into runoff hydrographs; (3) hydrodynamic simulation using a coupled 1D/2D HEC-RAS model to generate high-resolution flood inundation maps under a validated, compound-hazard scenario; and (4) a multi-criteria risk assessment that uses a k-means++ clustering algorithm to classify roadway segments into distinct risk hotspots based on their physical exposure, infrastructural characteristics, and socio-economic context. The complete workflow of this integrated framework is illustrated in Figure 2 used for identifying flood-prone roadway segments using hydrodynamic modeling and unsupervised clustering. The workflow integrates topographic, land use, meteorological, transportation, and socio-economic data to generate a flood inundation map and classify roadway risk hotspots.

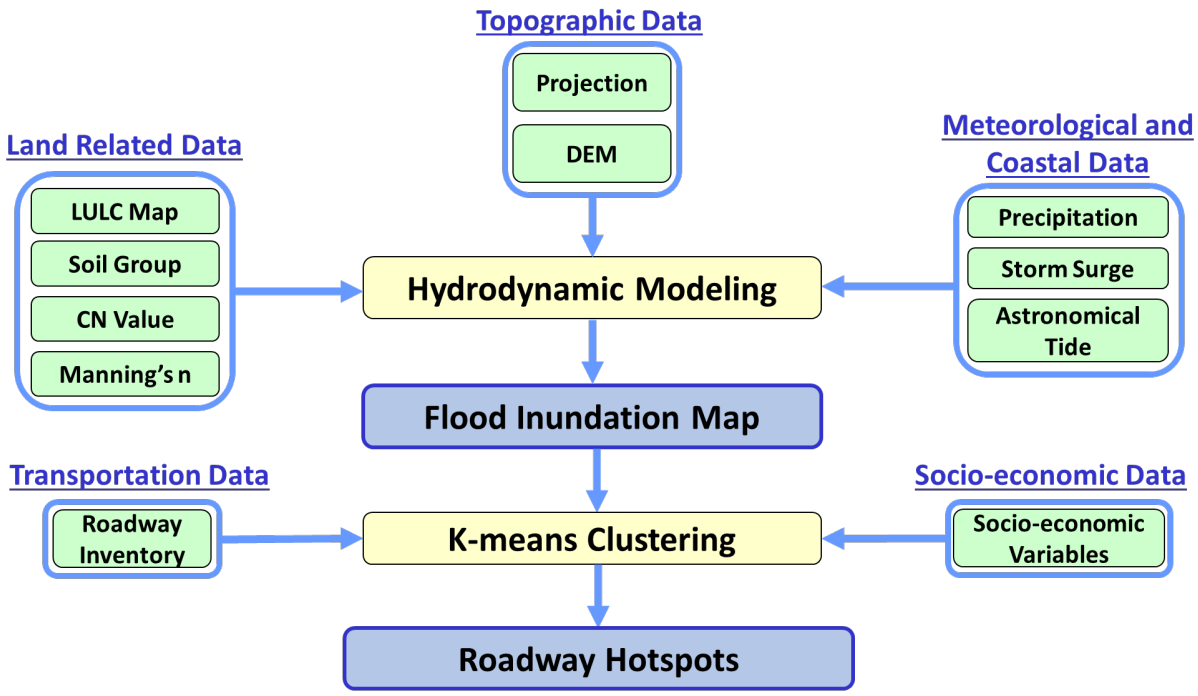

Figure 2. Integrated workflow for identifying roadway flood vulnerability hotspots. Topographic, land-use, soil, hydrologic, and coastal meteorological data are incorporated into hydrodynamic modeling to generate a flood inundation map. Roadway inventory attributes and socio-economic indicators are then combined with the modeled flood depths and analyzed using K-means clustering to classify roadway segments into distinct vulnerability hotspot categories.

2.2.1 Hydrodynamic Modeling

The flood hazard assessment at the core of this study was conducted using a coupled hydrologic-hydraulic modeling approach to simulate compound flooding dynamics across Galveston County. This process involved the development and validation of a high-resolution hydrodynamic model, which was then used to simulate a 100-year design storm event. All

modeling was performed using the U.S. Army Corps of Engineers' HEC-RAS (version 6.6) and HEC-HMS (version 4.12) software, supported by ArcGIS Pro and Python for data processing and analysis. The choice of a coupled 1D/2D HEC-RAS model was driven by its proven ability to represent complex flow patterns in both defined riverine channels (1D) and across broad, urbanized floodplains (2D) where flow is multi-directional and influenced by both fluvial and coastal drivers (Jibhakate et al., 2022; Pasquier et al., 2019; Shaikh et al., 2023). The model geometry was developed from a 3-meter resolution DEM from the USGS, as high-resolution topography is critical for accurately capturing the urban features and low-gradient terrain that governs floodwater propagation (Afshari et al., 2018; Purvis et al., 2008; Wadey et al., 2015; Yalcin, 2020). Surface roughness, a key parameter controlling flow velocity, was defined using spatially distributed Manning's *n* values derived from the 2023 NLCD data, a standard practice for enhancing the physical realism of 2D models (Afshari et al., 2018; Brunner, 2021; Di Salvo et al., 2018). The NLCD was also instrumental in defining spatially variable friction slopes for the downstream normal depth boundary conditions.

To ensure the model's reliability, it was first validated by hindcasting the flood event caused by Hurricane Ike, which made landfall on September 13, 2008. This specific 24-hour period was simulated as it corresponds to the time when the maximum storm tide was recorded at the Galveston Pier 21 station (Berg, 2009). The model was forced with a stage hydrograph representing the observed storm tide and excess rainfall derived from historical meteorological data. The simulated maximum flood extent and depths were compared against post-storm high-water marks documented by the FEMA. The practice of validating hydraulic models against observed HWMs is a well-established method for assessing model performance in real-world events (Barnard et al., 2014; Cook & Merwade, 2009; Teng et al., 2017). The comparison yielded a coefficient of determination (R^2) of 0.62, indicating a satisfactory performance consistent with the challenges of regional-scale modeling in complex coastal environments with multiple, interacting flood drivers (Dimitriadis et al., 2016; Green et al., 2025; Knebl et al., 2005).

Following successful validation, the model was used to simulate a 100-year compound flood scenario. This 24-hour simulation was designed to capture the combined impact of extreme inland rainfall and coastal storm surge, which is essential for a comprehensive risk assessment (Khanam et al., 2021; Shen et al., 2019; Wahl et al., 2015). The boundary conditions were defined as follows: the upstream forcing was derived from a design storm based on precipitation depths from NOAA Atlas 14, which provides the official precipitation frequency estimates for the United States (Perica et al., 2018). This rainfall was converted into excess rainfall hydrographs using the HEC-HMS model. The downstream coastal boundary was represented by the dynamic stage hydrograph from Hurricane Ike. The use of this hydrograph as a 100-year coastal event was justified by a flood frequency analysis of long-term water level data from NOAA, which confirmed that the Generalized Extreme Value (GEV) distribution provided the best fit and that the peak water levels during Ike were representative of a 100-year coastal flood. The GEV distribution is a standard and widely accepted statistical model for analyzing extreme sea levels and storm tides (Arns et al., 2013; Wahl et al., 2017; Zheng et al., 2013). The output from this simulation, a raster grid of maximum flood inundation depth, serves as the primary physical hazard input for the subsequent risk analysis.

2.2.2 Risk Assessment

Following the development of the flood hazard map, a comprehensive risk assessment was conducted to move beyond simple inundation mapping and classify roadway segments based on

their multifaceted risk profile. This process integrated the simulated flood hazard data with detailed infrastructure and socio-economic datasets, which were then analyzed using a sequence of statistical and machine learning techniques to identify and categorize roadway hotspots. To create a holistic risk profile for each road segment, the simulated 100-year flood depth and water velocity grid was spatially joined with the TxDOT 2023 roadway inventory and subsequently enriched with roadway characteristics (e.g., functional class, Average Annual Daily Traffic (AADT)) and socio-economic variables from the SLD data, thereby establishing a multi-criteria framework that recognizes risk as a tripartite function of exposure (flood depth and water velocity), sensitivity (physical and social characteristics), and adaptive capacity (network redundancy and mobility options) (Anand & Marcotullio, 2024; Clarke et al., 2024; Cutter et al., 2003; Fekete, 2011). By synthesizing the granular roadway attributes provided in the Texas Department of Transportation (TxDOT) Roadway Inventory with the block-group level socio-economic data of the SLD, this research facilitates a move beyond binary "flooded/not flooded" mapping toward a nuanced typology of risk.

Prior to classification, the integrated dataset underwent a rigorous pre-processing phase to ensure the validity and robustness of the subsequent analysis. First, variable formatting was standardized; specifically, zero-vehicle population counts were converted to population fractions to normalize for demographic density, and highly skewed distributions—including Traffic Volume (AADT), Total Population, and Employment—underwent logarithmic transformation to stabilize variance. A Risk Directionality Alignment was performed to ensure that for all variables, a higher numerical value consistently represented higher vulnerability. Specifically, variables such as functional classification of the roadways were inverted so that lower administrative codes (e.g., Interstates) were mathematically transformed to represent higher criticality scores. Following this alignment, a Pearson Correlation Matrix was computed to assess multicollinearity among the candidate variables. This diagnostic step validated the relationships between SLD parameters and hazard data, ensuring that redundant variables were not double-counted and confirming that the selected features represented distinct dimensions of vulnerability. Table 2 shows the transformation logic used for each dataset.

Table 2. Operationalization of vulnerability dimensions and risk directionality alignment protocols applied to input variables. The transformation rules normalize the directional influence of each variable so that higher values consistently represent higher flood-related risk across network segments.

Category	Variable	Source	Transformation Logic
Hazard	Max Flood Depth (m)	Flood Model	Min-Max Scaling. Normalized to 0–1 range (Higher = Greater Inundation Risk).
Hazard	Flood Velocity (m/s)	Flood Model	Min-Max Scaling. Normalized to 0–1 range (Higher = Greater Structural Stress).
Traffic	Traffic Volume (AADT)	TxDOT	Log-Transformation. Applied log(1+x) to reduce skewness of heavy-tailed distribution, then scaled.
Socio-economic	Total Population	SLD	Log-Transformation. Applied log(1+x) to normalize density variations between urban and rural segments.
Socio-economic	Total Employment	SLD	Log-Transformation. Applied log(1+x) to capture economic activity centers without outlier dominance.
Traffic	Truck % (Freight)	TxDOT	Min-Max Scaling. Normalized to 0–1 range (Higher = Critical Supply Chain Route).
Traffic	Functional Class	TxDOT	Inverted. Raw codes (1=Interstate, 7=Local) were mathematically inverted so that Interstates carry the highest numerical risk weight.
Traffic	Max Posted Speed	TxDOT	Min-Max Scaling. Used as a proxy for arterial importance (Higher Speed = Higher Criticality).
Socio-economic	Zero-Vehicle Population	SLD	Ratio Conversion. Converted from raw count to Fraction to normalize for density, then scaled.

To normalize the data ranges, all variables were scaled using Min-Max Normalization. This critical step rescales all features to a range of 0 to 1, ensuring that variables with large magnitudes, such as Traffic Volume, do not overpower variables with smaller ranges, such as Flood Depth, thereby preventing features with larger numeric ranges from dominating the clustering outcome. To reduce dimensionality and mitigate multicollinearity among input features, Principal Component Analysis was applied to the standardized dataset. The resulting principal components, which retained the majority of the data's variance, were subsequently used for clustering analysis (Abdi & Williams, 2010). Consistent with the pipeline settings, principal components explaining approximately 90% of the cumulative variance were retained for the final model.

To objectively classify the roadway segments, the k-means++ clustering algorithm was applied to the processed principal components. K-means++ is a powerful unsupervised machine learning method that partitions a dataset into a pre-defined number of clusters by iteratively minimizing the within-cluster variance, thereby grouping the most similar observations together (Jain, 2010; Steinley, 2006). This data-driven approach avoids the subjectivity inherent in manually defined risk thresholds (Javadi et al., 2017; Uzcategui-Salazar & Lillo, 2022) and has been successfully applied to identify risk patterns in urban flooding and transportation studies (Bhattacharyya et al., 2023; Bounaceur et al., 2022; Clarke et al., 2024). The K-Means++ initialization was specifically chosen to optimize the selection of initial centroids, ensuring faster

convergence and distinct separation between risk groups (Wu, 2012). The optimal number of clusters (k) was determined using a dual-validation approach combining the Elbow Method and Silhouette Analysis (Jain, 2010; Wu, 2012). The Elbow Method served as the primary decision criterion, revealing a sharp decrease in inertia at k=4, marking the distinct inflection point where marginal gains in compactness diminished. This selection was corroborated by the Silhouette Analysis, which yielded an average score of 0.386. While indicative of the complex, continuous nature of geospatial risk gradients, this score confirms a reasonable structural separation, establishing that the four-cluster solution offers the most mathematically robust and operationally distinct categorization of the network.

Finally, to assign semantic meaning to the mathematically derived clusters, a systematic post-hoc ranking methodology was applied. Rather than arbitrarily naming clusters, a composite vulnerability score was calculated for each cluster based on the proportion of segments exceeding defined critical thresholds: hydraulic hazard (depth ≥ 0.30 m or velocity ≥ 1.00 m/s), operational criticality (AADT ≥ 50,000 veh/day), and social vulnerability (zero-vehicle population ≥ 20%). This quantitative ranking process yielded the final risk typologies (Minor, Moderate, Major, Catastrophic), producing a spatially explicit map of roadway risk hotspots that provides a granular and evidence-based tool for targeted adaptation and emergency planning.

3. RESULTS

The analytical phase of this study yielded three sequential outputs that collectively form the basis for risk assessment. This section first presents the results of the hydrodynamic model validation, in which the HEC-RAS simulation of 2008 Hurricane Ike is quantitatively and qualitatively compared against observed data. Subsequently, the findings from the 100-year compound flood scenario are detailed, establishing the spatial extent and depth of the design flood hazard. Finally, the section concludes with the results of the k-means clustering analysis, which classifies the roadway network into distinct risk hotspots.

3.1 2008 Hurricane Ike Simulation and Model Validation

To establish the reliability of the hydrodynamic model, a hindcast of the Hurricane Ike flood event was performed for September 13, 2008. Hurricane Ike was a long-lived and expansive tropical cyclone that originated off the west coast of Africa, intensifying into a Category 4 hurricane in the Atlantic before traversing the Caribbean and entering the Gulf of Mexico (Berg, 2009). The system made landfall along the upper Texas coast near Galveston on September 13, 2008, as a Category 2 hurricane with sustained winds of 110 mph, driving a catastrophic storm surge that inundated significant portions of the Texas and Louisiana coastlines (*NOAA Water Level and Meteorological Data Report*, 2009).The developed HEC-RAS model produced a detailed map of maximum flood depths (Figure 3), which captured the key characteristics of the storm's impact. The simulation shows extensive inundation along the immediate coastline of both the islands with high water levels propagating deep into the Galveston Bay estuary system and its connected bayous. This spatial pattern is consistent with the known impacts of a major hurricane-induced storm tide in the region.

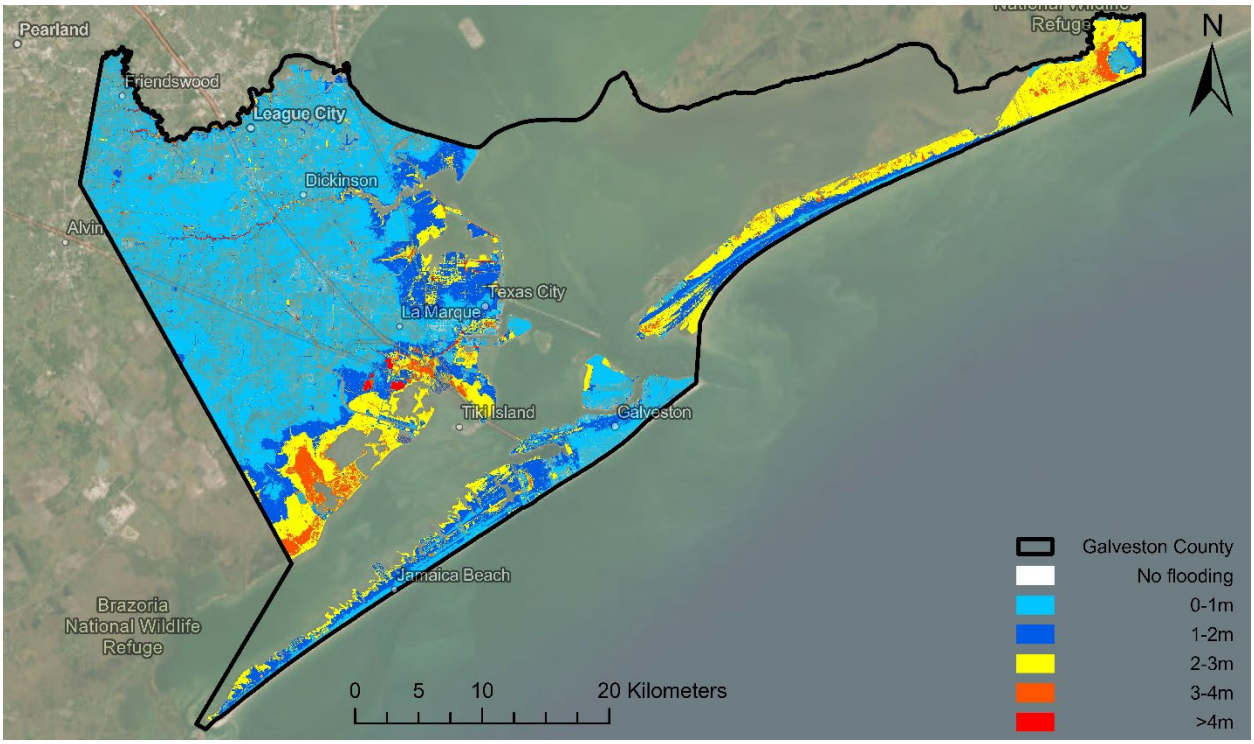

Figure 3. Spatial distribution of coastal flood depths across Galveston County derived from the hydrodynamic simulation of Hurricane Ike event. Flood depths are mapped using six hazard classes ranging from *no flooding* to *>4 m*, highlighting pronounced inundation along the eastern shoreline, Bolivar Peninsula, and low-lying communities in the central and southern coastal zones. The county boundary is overlaid to emphasize spatial variation of hazard intensity across the study area.

The quantitative performance of the coupled HEC-HMS and HEC-RAS model was evaluated by comparing simulated peak flood depths against observed High Water Marks (HWMs) collected by FEMA. As illustrated in Figure 4, the model demonstrates a strong ability to reproduce the spatial distribution of the flood event, evidenced by a Pearson correlation coefficient (r) of 0.78 and a coefficient of determination (R^2) of 0.62. These values indicate that the model explains approximately 62% of the variance in the observed flood depths, suggesting that the underlying hydraulic connectivity and terrain schematization in the 2D domain are functioning correctly. The error analysis yielded a Mean Absolute Error (MAE) of 0.46 m and a Root Mean Squared Error (RMSE) of 0.64 m. A systematic tendency toward underestimation was observed, indicated by the Mean Bias Error (MBE) of -0.30 m and the prevalence of negative residuals in the error plot. This bias likely reflects the distinction between the static mean water levels simulated by the hydrodynamic model and the dynamic nature of HWMs, which often capture "mud lines" resulting from wave run-up and superelevation.

Regarding model efficiency, the Nash-Sutcliffe Efficiency (NSE) was 0.37, while the log-transformed NSE (log-NSE) improved to 0.55. Furthermore, the model achieved a Kling–Gupta Efficiency (KGE) of 0.66, which indicates a good balance between correlation, bias, and variability. The divergence between the moderate NSE and the stronger correlation (r=0.78) suggests that while the model captures the overall hydraulic patterns and flow routing effectively, it remains conservative in replicating absolute peak magnitudes at specific locations. Despite this

conservative bias, the validation results confirm the model's utility for comparative scenario analysis. The high correlation demonstrates systematic consistency—the model correctly identifies topographic drivers of flooding. Consequently, when simulating future scenarios, the relative difference between the baseline and future conditions remains robust, as the systematic bias effectively cancels out in the comparison. These metrics fall within the reasonable range for complex, regional-scale coastal models, supporting the model's application for risk assessment (Dimitriadis et al., 2016; Knebl et al., 2005).

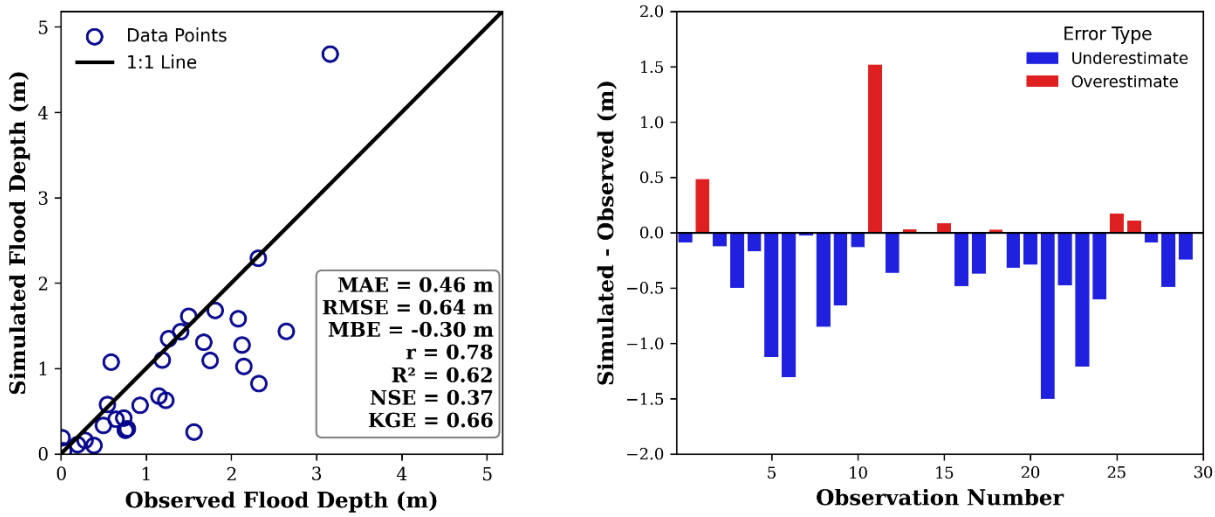

Figure 4. Comparison of observed and simulated flood depths. The left panel presents a scatter plot of the simulated versus observed flood depths relative to the 1:1 line of perfect agreement, including statistical performance metrics: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Bias Error (MBE), Pearson correlation coefficient (r), coefficient of determination (R^2), Nash–Sutcliffe Efficiency (NSE), and Kling–Gupta Efficiency (KGE). The right panel illustrates the error residuals (Simulated minus Observed) for individual observation points, distinguishing between model underestimations (blue bars) and overestimations (red bars).

3.2 100-Year Design Flood Simulation

Following the successful model validation, HEC-RAS model was used to simulate the 100-year, 24-hour compound flood event. The resulting map of maximum flood inundation depth, which represents the primary physical hazard for the subsequent risk assessment, is presented in Figure 5. From the figure it can be depicted that depths are classified into four categories to support spatial assessment of flood-prone areas. High inundation zones are primarily concentrated along the coast and low-lying inland areas, providing critical input for transportation risk analysis.

The simulation reveals a widespread and severe flood hazard across Galveston County. The spatial distribution of inundation clearly illustrates the compound nature of the event. The most extreme flood depths, in many areas around 2.0 meters, are concentrated along the immediate coastline of Galveston Island, the Bolivar Peninsula, and the Galveston Bay shoreline, reflecting the dominant impact of the 100-year storm tide. The model shows significant inland propagation of these high-water levels through the estuary system and into connected bayous. Simultaneously, the elevated coastal water levels impede the drainage of runoff generated by the 100-year rainfall event, causing significant backwater effects. This leads to extensive, deep flooding along inland bayous and in

low-lying terrestrial areas that might otherwise be unaffected by a coastal surge alone. The resulting inundation poses a significant threat to the county's infrastructure, with numerous major transportation corridors shown to be inundated by substantial water depths. This detailed hazard map serves as the foundational input for the risk analysis of the roadway network.

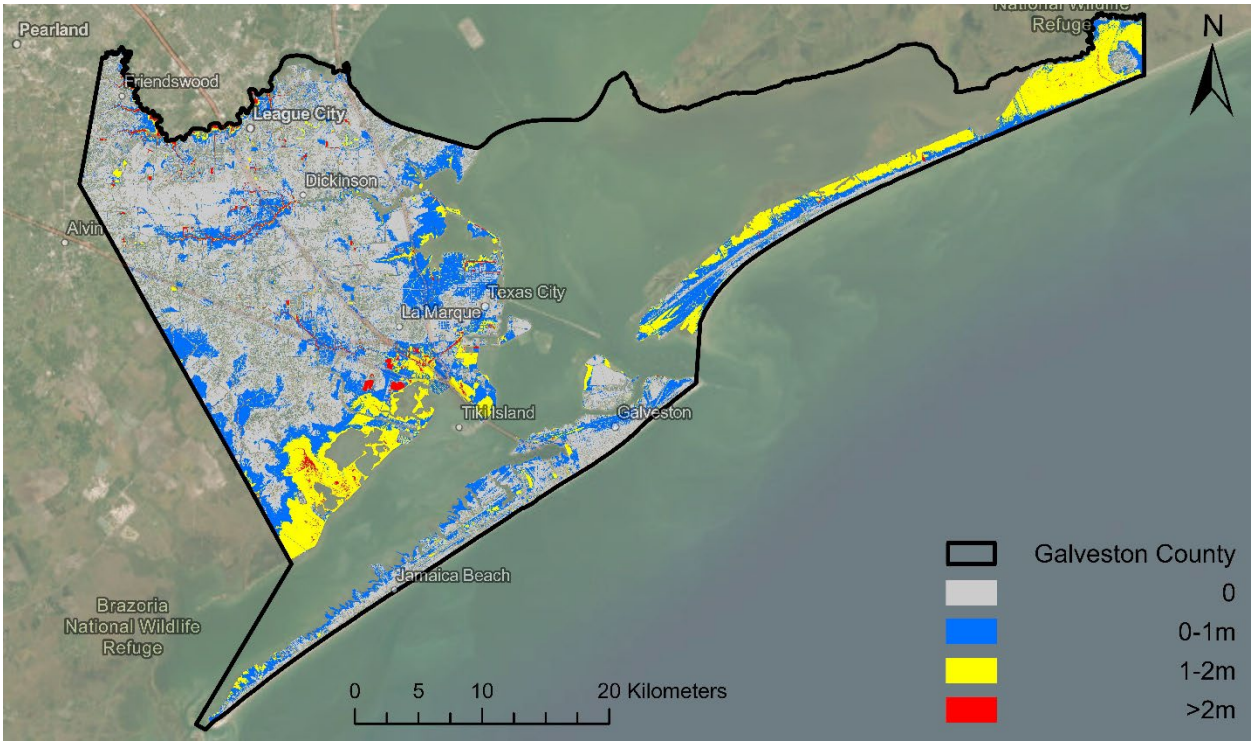

Figure 5. Spatial distribution of coastal flood depths across Galveston County for the 100-year return period event. Flood intensity is classified into four depth categories (0, 0–1 m, 1–2 m, and >2 m), highlighting substantial inundation along the Bolivar Peninsula, eastern shoreline, and low-lying communities within central and southern coastal zones. The county boundary is overlaid to illustrate spatial variation of flood severity across the study area.

3.3 Roadway Network Risk Classification

This study utilized K-Means++ clustering to systematically identify and classify vulnerable roadway hotspots across the network. To determine the optimal granularity for this classification, the Elbow Method was applied as the primary decision criterion, revealing a decisive inflection point at $k=4$ where gains in cluster compactness plateaued. This configuration was further validated by a Silhouette score of 0.386, confirming that a four-cluster solution offers the most mathematically robust and operationally distinct partition of the risk landscape. The K-Means++ clustering algorithm successfully partitioned the roadway segments within the study area into four statistically distinct risk typologies based on a multi-dimensional assessment of hydraulic exposure (depth, velocity), operational criticality (AADT), and social vulnerability (zero-vehicle ownership). The spatial distribution of these categories is illustrated in Figure 6. Unlike simple inundation maps, this typology differentiates segments based on the *mechanism* of vulnerability—whether the risk is driven by infrastructure importance, physical hazard intensity, or the demographic sensitivity of the population served.

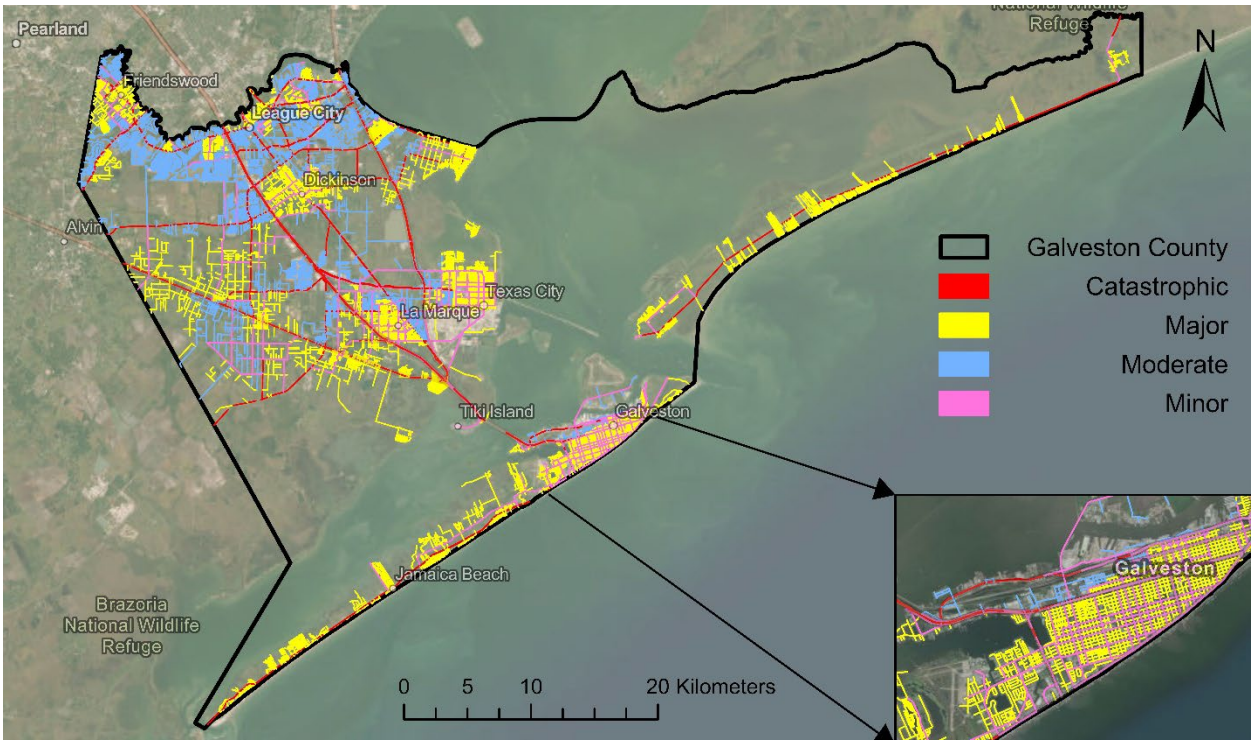

Figure 6. Spatial distribution of identified roadway vulnerability typologies across Galveston County. The roadway network is partitioned into four distinct risk clusters—Minor (Pink), Moderate (Blue), Major (Yellow), and Catastrophic (Red)—derived from the K-Means++ clustering analysis. The inset highlights the dense urban grid of Galveston Island, revealing the concentration of Catastrophic risk along primary evacuation corridors (e.g., I-45 Causeway) and Major structural risk within the urban core.

Based on the cluster profiles, the four classes describe a spectrum of vulnerability ranging from social isolation to operational failure. The Minor Vulnerability (Pink) category characterizes local access roads in socially vulnerable neighborhoods; while these segments exhibit low physical exposure (Figure 7a), they contain the highest concentration of zero-vehicle households (Figure 7g), making "shelter-in-place" isolation the primary risk mechanism rather than structural failure. In contrast, the Moderate Risk (Blue) cluster encompasses the suburban "background" of the network, where high residential density and vehicle ownership rates coincide with minimal flood depths to create a resilient typology requiring only post-event debris clearance. Moving toward greater severity, the Major Risk (Yellow) category represents a "double jeopardy" zone within the dense urban grid, distinguishing itself from the arterial network by combining extreme inundation depths comparable to the highest risk tier with high population transit dependency. Finally, the Catastrophic Failure (Red) cluster isolates the region's logistical backbone—such as the I-45 Causeway and State Highway 87—which is defined uniquely by operational intensity, carrying the highest traffic volumes, posted speeds, and truck shares essential for mass evacuation and supply chain continuity.

The statistical distinctiveness of these four risk classes is validated by the violin plots presented in Figure 7, which confirm that the clusters represent physically and operationally distinct realities rather than arbitrary partitions. Regarding physical hazard, the Catastrophic and Major clusters are clearly isolated by extreme flood depths (>2 m) and velocities, distinguishing

them from the nuisance-level flooding of the blue and pink clusters. Operationally, the Catastrophic cluster remains statistically unique; it is the only group with a median posted speed exceeding 80 km/h (Panel e) and significant freight utility (Panel d), confirming the algorithm correctly prioritized economic lifelines over local streets. Furthermore, the analysis reveals that social vulnerability is the primary driver for the Minor and Major clusters, both of which show extended probability tails for zero-vehicle populations.

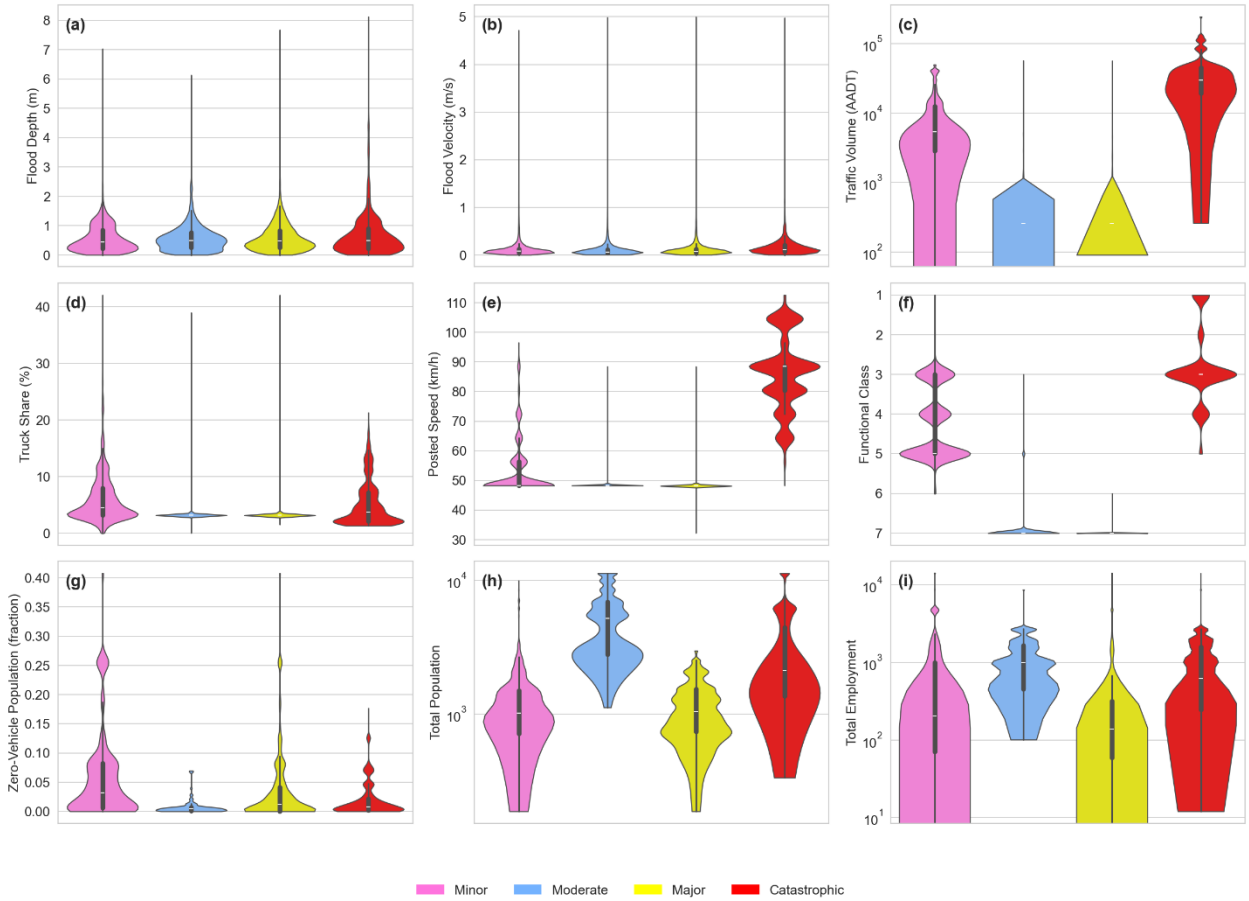

Figure 7. Distribution of key hazard, exposure, and social vulnerability indicators across flood severity classes (Minor, Moderate, Major, and Catastrophic). *(a)* Flood depth distributions, *(b)* flood velocity distributions, *(c)* traffic volume (AADT) distributions shown on a logarithmic scale, and *(d)* proportion of zero-vehicle households. Violin plots illustrate the full distribution and variability within each class, with embedded boxplots indicating median and interquartile ranges. The results highlight systematic increases in hazard intensity and exposure with flood severity, alongside distinct differences in traffic demand and social vulnerability characteristics across classes.

The robustness of this classification is further corroborated by the Principal Component Analysis in Figure 8. The loading matrix reveals that the variance structure of the roadway network is primarily defined by socio-economic geography rather than physical hazard alone; the first principal component (PC1), explaining 55% of the variance, is heavily loaded by Social Vulnerability, while the third component (PC3) captures Economic Intensity. This orthogonality underscores the strength of the clustering logic, where the Minor cluster separates distinctively along the PC1 axis to identify socially vulnerable segments lacking economic intensity.

Conversely, the Catastrophic cluster diverges sharply along the PC3 axis, effectively isolating the high-traffic, high-employment arterial corridors. Lying between these extremes, the Major cluster occupies a unique centroid defined by the intersection of high social vulnerability (PC1) and high physical hazard, validating its classification as a critical zone of compounding risk.

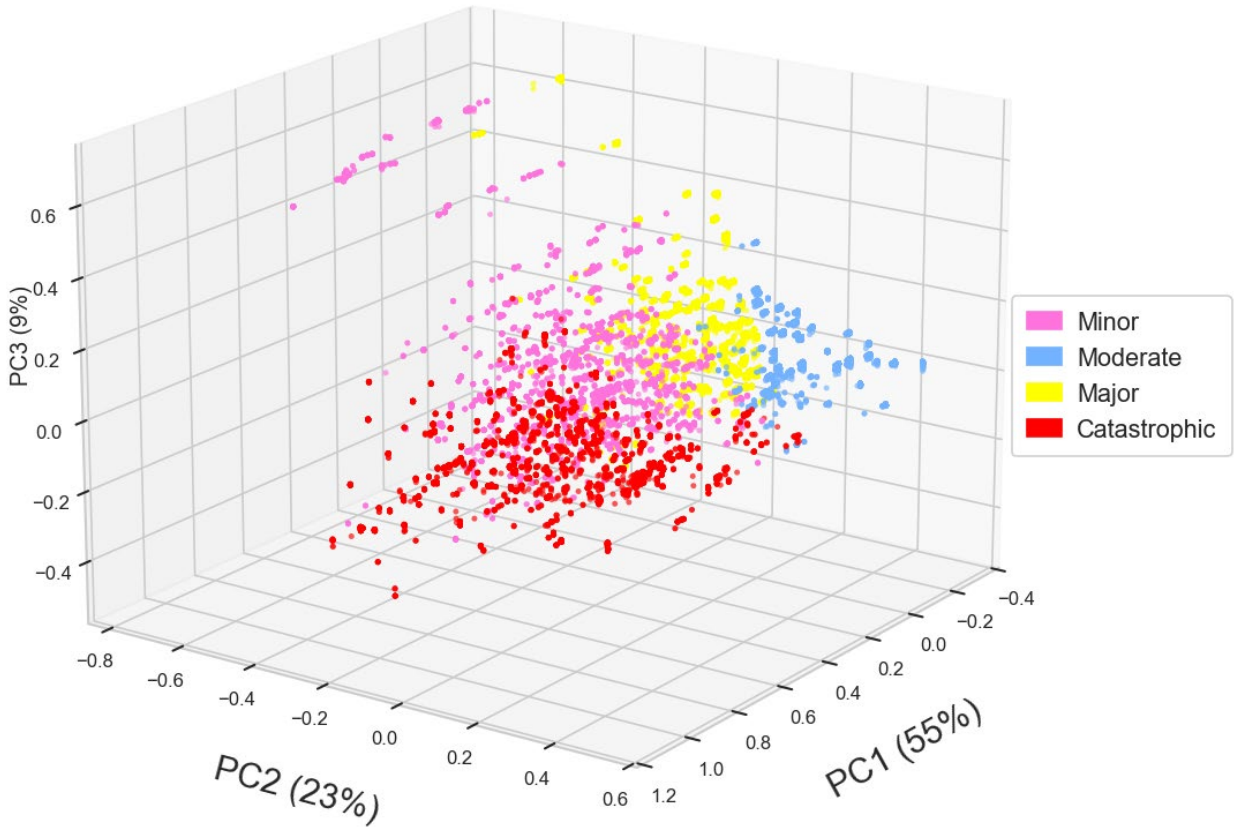

Figure 8. Three-dimensional principal component analysis (PCA) of roadway segments colored by flood vulnerability class derived from K-means clustering. The first three principal components explain 87% of the total variance (PC1 = 55%, PC2 = 23%, PC3 = 9%). Distinct separation is observed across the four severity classes (Minor, Moderate, Major, and Catastrophic), indicating robust multivariate differentiation driven by flood hazard, traffic criticality, and socio-economic exposure. Clustering reveals increasing concentration of high-risk segments along PC1 and PC2, consistent with monotonic increases in inundation depth, traffic volume, and social vulnerability.

4. DISCUSSION

The results of this study provide critical insights into the compound flood risk facing Galveston County's transportation network and offer a robust, data-driven basis for enhancing coastal resilience. By coupling high-resolution hydrodynamic modeling with unsupervised machine learning, the findings demonstrate the necessity of modeling compound events to capture

the full extent of hazard exposure, and the value of a multi-criteria approach to move beyond simple inundation mapping toward a holistic assessment of systemic risk.

The simulation of the 100-year compound event confirms that the flood hazard in Galveston County is not a simple summation of independent coastal and riverine threats. Relying on traditional floodplain maps that do not account for these compound dynamics can lead to a significant underestimation of risk, leaving communities and critical infrastructure unwittingly exposed (Ray et al., 2011). Our results specifically illustrate a dramatic increase in inundated areas when cyclone-induced rainfall is included, reinforcing that precipitation is a dominant driver in compound events rather than a secondary factor (Pandey et al., 2021). The resulting inundation map, therefore, represents a more realistic threat assessment, particularly for infrastructure that appears safe on single-hazard maps but is vulnerable to compound effects (Khanam et al., 2021).

However, the risk classification analysis (Figure 6) highlights that physical exposure is only one dimension of vulnerability. As revealed by the PCA and clustering results, a roadway segment's risk is significantly amplified by its functional importance and the socio-economic context of the area it serves. For instance, the identification of the "Catastrophic Failure" cluster isolates the I-45 Causeway and State Highway 87—routes defined not just by extreme flood depths, but by their role as the region's operational backbone. Its failure could lead to severe "knock-on" effects, isolating entire communities and impeding emergency response, a key concern in infrastructure resilience (Pyatkova et al., 2019; Uzcategui-Salazar & Lillo, 2022). In contrast to the arterial spine, the "Major Risk" cluster captures the dense urban grid where high physical hazard intersects with high social vulnerability. This cluster represents a "double jeopardy" scenario where flooded collector roads trap transit-dependent populations. Similarly, the identification of risk roads providing sole access to hospitals or other critical facilities underscores the systemic nature of flood risk, where the disruption of one asset can cascade through the community (Borowska-Stefańska et al., 2025; Najafi et al., 2021).

This study's methodology provides a tangible tool for transportation agencies and emergency managers to prioritize adaptation investments. Instead of a one-size-fits-all approach based solely on flood depth, the generated risk map allows for a targeted allocation of resources. For example, "Catastrophic" hotspots on critical arteries may warrant significant capital investment in structural protection or elevation to maintain regional connectivity (Mouronte-López, 2021). Conversely, areas classified as "Minor Vulnerability" or "Moderate Risk" —which face lower structural threats but serve specific community needs might be better served by community-scale solutions, such as enhanced local drainage or green infrastructure, to reduce runoff and provide co-benefits (Liu et al., 2022; Rajput et al., 2022). This risk-based prioritization aligns with modern frameworks that seek to balance economic efficiency with social equity (Martello et al., 2021; Mitropoulos et al., 2025). By explicitly incorporating socio-economic factors like zero-car households into the clustering algorithm, this approach ensures that adaptation planning addresses the needs of the most vulnerable populations, a critical aspect of equitable resilience (Clarke et al., 2024).

5. LIMITATIONS AND FUTURE WORK

While this framework provides a robust and replicable assessment of roadway vulnerability, several limitations inherent to data availability and study design must be acknowledged. A primary constraint lies in the meteorological and boundary condition inputs used for the hydrodynamic modeling. Due to the limited availability of long-term historical records at alternative sites, this study relied on a single NOAA tide gauge to define the coastal boundary

condition and a single precipitation station to characterize rainfall across the entire county. Consequently, the model assumes spatially uniform rainfall and surge levels, which, while necessary due to data unavailability, does not fully capture the practical spatial variability of storm dynamics across a region as large as Galveston County. Additionally, the hydrodynamic model, despite its validation, does not explicitly account for wind-driven wave action or localized wave run-up, which can exacerbate flooding in immediate coastal zones.

Methodologically, the risk classification is further constrained by the spatial resolution of the input datasets and the static nature of the analysis. Regarding the socio-economic integration, attributes were derived from the Smart Location Database at the census block group level; consequently, all road segments within a single block group were assigned identical socio-economic values. This generalization masks hyper-local variations in social vulnerability that might exist between different streets within the same block. Furthermore, the risk analysis remains a static snapshot of vulnerability under a single 100-year design storm. It does not capture the full spectrum of risk from more frequent, lower-magnitude events, nor does it model the dynamic, time-dependent nature of traffic disruption and network-wide cascading failures that occur as a flood event evolves.

Future work should aim to address these limitations by incorporating higher-resolution spatial data and dynamic modeling techniques. To overcome the data scarcity issues, future studies could utilize spatially distributed radar-derived rainfall data and multi-point coastal observations to better represent the heterogeneity of compound hazards. To address the static nature of the current assessment, researchers should integrate dynamic traffic simulation to quantify the economic costs of delays and evaluating network redundancy. Finally, extending the analysis to include future climate change and sea-level rise scenarios would provide a more comprehensive view of long-term infrastructure resilience.

6. CONCLUSION

This study successfully developed and applied an integrated framework for assessing the resilience of coastal transportation networks to compound flooding, utilizing Galveston County, Texas, as a critical testbed. By coupling a validated high-resolution 1D/2D hydrodynamic model with an unsupervised machine learning classification, this research advances risk assessment beyond binary hazard mapping toward a holistic quantification of systemic vulnerability. The findings underscore that coastal infrastructure resilience is not merely a function of physical robustness but a complex interaction of hazard exposure, operational criticality, and social equity.

The hydrodynamic simulation of the 100-year compound event confirmed that the interaction of storm surge and heavy rainfall creates a hazard significantly more severe than the summation of isolated drivers. The model captured extensive backwater effects where elevated bay levels prevented inland drainage, inundating critical infrastructure that would otherwise appear safe in single-hazard assessments. However, the subsequent risk analysis revealed that roadway vulnerability is not linearly dependent on this flood depth. Instead, as demonstrated by the Principal Component Analysis, the variance in network vulnerability is driven as much by socio-economic geography as by physical hazard.

The machine learning classification effectively partitioned the network into distinct typologies that demand differentiated adaptation strategies. The analysis identified a "Catastrophic Failure" tier representing the region's logistical spine, where extreme hydraulic stress converges with high operational intensity, necessitating structural hardening. Conversely, it isolated a "Major Risk" tier within the dense urban grid, representing a "double jeopardy" zone where severe

inundation intersects with high transit dependency. By objectively distinguishing these critical nodes from "Minor" and "Moderate" vulnerability hotspots—where the primary threat is isolation rather than structural failure—this framework offers a tangible tool for decision-makers. It enables a targeted allocation of resources that balances the protection of the physical economy with the equitable preservation of community mobility.

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8. AUTHOR CONTRIBUTIONS

The authors confirm contribution to the paper as follows: Md Shah Mominul Islam Momin: conceptualization, methodology, draft and final manuscript preparation, and result visualization; Eunsang Cho: project supervision, conceptualization, and manuscript review/editing; Eunsaem Cho: methodology and manuscript review/editing; Subasish Das: conceptualization, data guidance, and manuscript review/editing. All authors reviewed the results and approved the final version of the manuscript.

9. DECLARATION OF CONFLICTING INTERESTS

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

10. DATA AVAILABILITY STATEMENT

The data that support the findings of this study will be made openly available in [Zenodo] upon acceptance of this manuscript. The repository will include the source inputs, specifically the shapefiles and raster files used for the flood modeling. Additionally, it will contain the calibrated simulation files, comprising the HEC-RAS geometry and the roadway network shapefile. Finally, the Python scripts used for coupling the physical, traffic, and socio-economic characteristics and generating the figures will be included. A permanent Digital Object Identifier (DOI) will be provided in this section prior to final publication.

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